

SECURITY OPERATIONS LAB

FILE INTEGRITY MONITORING POC

PREPARED BY:

Rohan Teja

July 2026

INTRODUCTION

This report documents a proof-of-concept exercise setting up File Integrity Monitoring (FIM) on a Windows endpoint using Wazuh's syscheck module, then testing whether it actually caught what it should: file creates, deletes, edits, and registry changes, plus who made the change and what changed in the file itself.

The report covers configuration, basic detection testing, alert and compliance-mapping detail, who-data mode for forensic attribution, and the content diff and MITRE auto-tagging Wazuh produced. It also documents the troubleshooting involved, rather than presenting the result as clean on the first try.

The key focus areas of this report:

1. Configuring FIM – setting the monitored paths, registry keys, and scan mode.
2. Basic Detection – confirming file add/delete/modify events are caught.
3. Alert Detail & Compliance Mapping – checking what context Wazuh attaches automatically.
4. Who-Data Mode – getting user and process attribution, not just "something changed."
5. Content Diff & MITRE Auto-Tagging – confirming exact before/after content and technique mapping.
6. Troubleshooting – the real configuration issues hit along the way.

Lab Environment

Machine	Role	Operating System	Purpose
Wazuh Manager	SIEM	Ubuntu Server (192.168.142.129)	Central log correlation, dashboard on port 443
DESKTOP-TDOGRJA	Monitored Endpoint	Windows 10 (192.168.142.130)	Agent 001; FIM target for file and registry monitoring
Kali Linux	Third VM	Kali Linux	Run alongside to simulate an attacker-present network segment

1. CONFIGURING FIM

FIM runs as Wazuh's syscheck module on the Windows agent. The default configuration already covered system paths, PowerShell directories, and the Startup folder; a custom test directory was added for this lab.

Windows Agent Configuration

Edit the Wazuh agent configuration file:

```
C:\Program Files (x86)\ossec-agent\ossec.conf
```

Add the test directory within the <syscheck> block:

```
<syscheck>
  <directories check_all="yes" report_changes="yes"
    realtime="yes">C:\Users\Public\Pictures</directories>
  <frequency>10</frequency>
</syscheck>
```

Restart the Wazuh agent via PowerShell as Administrator:

```
Restart-Service -Name wazuh
```

```

<!-- Frequency that syscheck is executed default every 12 hours -->
<frequency>18</frequency>

<!-- Default files to be monitored. -->
<directories recursion_level="0" restrict="regedit.exe|system.ini|win.ini">%WINDIR%</directories>
<directories recursion_level="1">C:\Users\Public</directories>
<directories recursion_level="0" restrict="at.exe|attrib.exe|cacls.exe|cmd.exe|eventcreate.exe|ftp.exe|lsass.exe|net.exe|net1.exe|netsh.exe|re
<directories recursion_level="0">%WINDIR%\SysNative\drivers\etc</directories>
<directories recursion_level="0" restrict="WMIC.exe">%WINDIR%\SysNative\wbem</directories>
<directories recursion_level="0" restrict="powershell.exe">%WINDIR%\System32\WindowsPowerShell\v1.0</directories>
<directories recursion_level="0" restrict="winrm.vbs">%WINDIR%\SysNative</directories>

<!-- 32-bit programs. -->
<directories recursion_level="0" restrict="at.exe|attrib.exe|cacls.exe|cmd.exe|eventcreate.exe|ftp.exe|lsass.exe|net.exe|net1.exe|netsh.exe|re
<directories recursion_level="0">%WINDIR%\System32\drivers\etc</directories>
<directories recursion_level="0" restrict="WMIC.exe">%WINDIR%\System32\wbem</directories>
<directories recursion_level="0" restrict="powershell.exe">%WINDIR%\System32\WindowsPowerShell\v1.0</directories>
<directories recursion_level="0" restrict="winrm.vbs">%WINDIR%\System32</directories>

<directories realtime="yes">%PROGRAMDATA%\Microsoft\Windows\Start Menu\Programs\Startup</directories>

<ignore>%PROGRAMDATA%\Microsoft\Windows\Start Menu\Programs\Startup\desktop.ini</ignore>

<ignore type="sregex">.log|.htm|.jpg|.png|.chm|.pnf|.evtx</ignore>

```

Figure 1 — ossec.conf: monitored directories, restrict rules, and registry keys on the Windows agent.

Scan interval was dropped to 10 seconds for testing (the default is 12 hours), with realtime monitoring on the Startup folder and ignore rules for noisy file types (.log, .jpg, .png, .evtx).

2. BASIC DETECTION TEST

To confirm detection worked, a file named malware-exe was created in the monitored folder.

Testing the Configuration

1. Create a test file in C:\Users\Public\Pictures.
2. Rename, edit, or delete it.
3. Wait 5–10 seconds for the alert to generate.

Viewing Alerts in the Wazuh Dashboard

1. Log in to the Wazuh dashboard.
2. Navigate to Security Events > File Integrity Monitoring.
3. Filter using: rule.groups:"syscheck" AND agent.name:"DESKTOP-TDOGRJA"

timestamp	agent.name	syscheck.path	syscheck.event	rule.description	rule ID
Jul 17, 2026 @ 23:20:15.969	DESKTOP-TDOGRJA	c:\users\public\pictures\new text document.txt	deleted	File deleted.	7
Jul 17, 2026 @ 23:20:15.327	DESKTOP-TDOGRJA	c:\users\public\pictures\malware-exe.txt	added	File added to the system.	5
Jul 17, 2026 @ 23:20:01.273	DESKTOP-TDOGRJA	HKEY_LOCAL_MACHINE\System\CurrentControlSet\Ser...	modified	Registry Value Integrity Checksum Changed	5
Jul 17, 2026 @ 23:20:01.231	DESKTOP-TDOGRJA	HKEY_LOCAL_MACHINE\System\CurrentControl\...	modified	Registry Value Integrity Checksum Changed	5
Jul 17, 2026 @ 23:20:01.216	DESKTOP-TDOGRJA	HKEY_LOCAL_MACHINE\System\CurrentControlSet\Ser...	modified	Registry Value Integrity Checksum Changed	5
Jul 17, 2026 @ 23:20:01.216	DESKTOP-TDOGRJA	HKEY_LOCAL_MACHINE\System\CurrentControlSet\Ser...	modified	Registry Key Integrity Checksum Changed	5
Jul 17, 2026 @ 23:20:01.215	DESKTOP-TDOGRJA	HKEY_LOCAL_MACHINE\System\CurrentControlSet\Ser...	modified	Registry Value Integrity Checksum Changed	5
Jul 17, 2026 @ 23:20:01.194	DESKTOP-TDOGRJA	HKEY_LOCAL_MACHINE\System\CurrentControlSet\Ser...	modified	Registry Key Integrity Checksum Changed	5
Jul 17, 2026 @ 23:19:34.490	DESKTOP-TDOGRJA	c:\users\public\pictures\malware.txt	deleted	File deleted.	7
Jul 17, 2026 @ 23:19:33.782	DESKTOP-TDOGRJA	c:\users\public\pictures\new text document.txt	added	File added to the system.	5
Jul 17, 2026 @ 23:15:16.806	DESKTOP-TDOGRJA	HKEY_LOCAL_MACHINE\System\CurrentControlSet\Ser...	modified	Registry Value Integrity Checksum Changed	5
Jul 17, 2026 @ 23:15:16.788	DESKTOP-TDOGRJA	HKEY_LOCAL_MACHINE\System\CurrentControlSet\Ser...	modified	Registry Value Integrity Checksum Changed	5

Figure 2 — Wazuh FIM event list: add/delete/modify events for files and registry keys, with rule ID and level.

The file showed up on the dashboard within seconds, alongside file-deleted and registry-checksum-changed events: 40 hits for the day across file and registry activity.

3. ALERT DETAIL & COMPLIANCE MAPPING

Opening the file-added alert (rule 554, level 5) showed the full picture: agent, path, decoder, and automatic mappings to GDPR, HIPAA, NIST 800-53, PCI-DSS, and TSC controls, attached without any manual tagging.

Field	Value
_index	wazuh-alerts-4.x-2026.07.17
agent.id	001
agent.ip	192.168.142.130
agent.name	DESKTOP-TDOGRJA
decoder.name	syscheck_new_entry
full_log	File 'c:\users\public\pictures\malware-exe.txt' added Mode: scheduled
id	1784310615.4735884
input.type	log
location	syscheck
manager.name	wazuh.manager
rule.description	File added to the system.
rule.firedtimes	2
rule.gdpr	II.5.1.f
rule.gpg13	4.11
rule.groups	ossec, syscheck, syscheck_entry_added, syscheck_file
rule.hipaa	164.312.c.1, 164.312.c.2
rule.id	554
rule.level	5
rule.mail	false
rule.nist_800_53	SI.7

Figure 3 — Alert detail for the malware-exe file-added event, with compliance references auto-attached by Wazuh.

4. WHO-DATA MODE — ATTRIBUTION

Scheduled and realtime mode tell you something changed. Neither tells you who. The test file was switched to who-data mode and edited in Notepad.

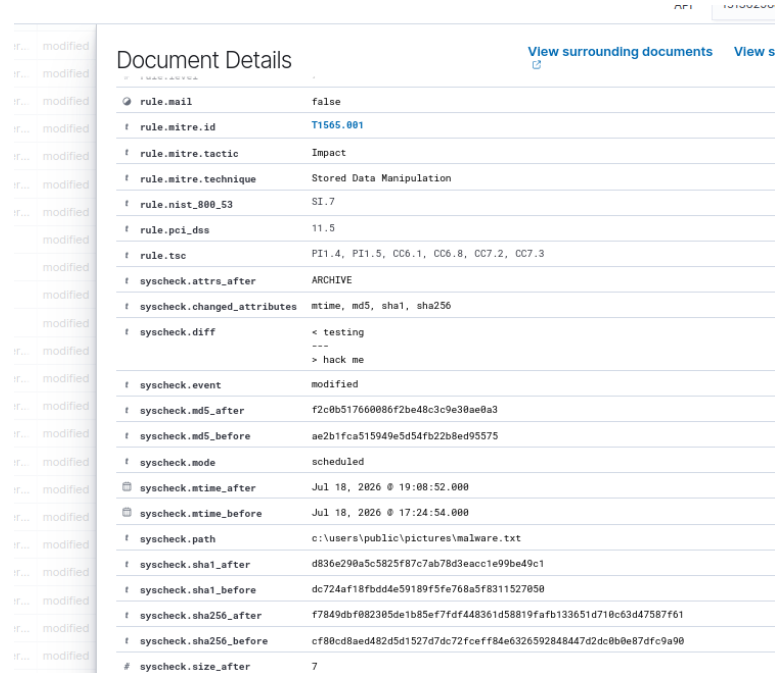
Field	Value
rule.mitre.technique	Stored Data Manipulation
rule.nist_800_53	SI.7
rule.pci_dss	11.5
rule.tsc	PI1.4, PI1.5, CO6.1, CO6.8, CO7.2, CO7.3
syscheck.attrs_after	ARCHIVE
syscheck.audit.process.id	7516
syscheck.audit.process.name	C:\Windows\System32\notepad.exe
syscheck.audit.user.id	S-1-5-21-3806784903-2991345533-3997175912-1001
syscheck.audit.user.name	Victim
syscheck.changed_attributes	size, mtime, md5, sha1, sha256
syscheck.event	modified
syscheck.md5_after	ae2b1fca515949e5d54fb22b8ed95575
syscheck.md5_before	d41d8cd98f00b204e9800998ecf8427e
syscheck.mode	whodata
syscheck.mtime_after	Jul 18, 2026 @ 17:24:54.000
syscheck.mtime_before	Jul 18, 2026 @ 17:24:44.000
syscheck.path	c:\users\public\pictures\malware.txt
syscheck.sha1_after	dc724af18fbd4e59189f5fe768a5f8311527050
syscheck.sha1_before	da39a3ee5e6b4b0d3255bfef95601890afd80709
syscheck.sha256_after	cf80cd8aed482d5d1527d7dc72fcef84e6326592848447d2dc0b0e87dfc9a90
syscheck.sha256_before	e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855
syscheck.size_after	7
syscheck.size_before	0
syscheck.uid_after	S-1-5-21-3806784903-2991345533-3997175912-1001
syscheck.uname_after	Victim

Figure 4 — Who-data alert: responsible user, process, and the MITRE technique Wazuh auto-tagged (Stored Data Manipulation).

The resulting alert included the user account (Victim), the exact process that wrote to the file (notepad.exe, PID 7516), and the full set of changed attributes (size, mtime, md5, sha1, sha256) with before/after values for each.

5. CONTENT DIFF & MITRE AUTO-TAGGING

Wazuh tagged the change as MITRE ATT&CK T1565.001 (Stored Data Manipulation) on its own, no manual technique lookup needed.



Document Details	
rule.mail	false
rule.mitre.id	T1565.001
rule.mitre.tactic	Impact
rule.mitre.technique	Stored Data Manipulation
rule.nist_800_53	SI.7
rule.pci_dss	11.5
rule.tsc	PI1.4, PI1.5, CC6.1, CC6.8, CC7.2, CC7.3
syscheck.attr_after	ARCHIVE
syscheck.changed_attributes	mtime, md5, sha1, sha256
syscheck.diff	< testing --- > hack me
syscheck.event	modified
syscheck.md5_after	f2c0b517660086f2be48c3c9e30ae8a3
syscheck.md5_before	ae2b1fca515949e5d54fb22b8ed95575
syscheck.mode	scheduled
syscheck.mtime_after	Jul 18, 2026 @ 19:08:52.000
syscheck.mtime_before	Jul 18, 2026 @ 17:24:54.000
syscheck.path	c:\users\public\pictures\malware.txt
syscheck.sha1_after	d836e290a5c5825f87c7ab78d3eacc1e99be49c1
syscheck.sha1_before	dc724af18fdd4e59189f5fe768a5f8311527850
syscheck.sha256_after	f7849dbf082305de1b85ef7f448361d58819fab133651d710c63d47587f61
syscheck.sha256_before	cf80cd8aed482d5d1527d7dc72f0eff84e6326592848447d2dc0b0e87dfc9a90
syscheck.size_after	7

Figure 5 — Second modification: MITRE T1565.001 re-confirmed, plus the exact before/after content diff and hashes.

Editing the file a second time ("testing" to "hack me") reproduced the same tag and also surfaced the literal content diff, syscheck.diff: < testing / > hack me, next to the updated hashes and the file's Windows ACL permissions.

MITRE ATT&CK Mapping

Only T1565.001 was directly auto-tagged by Wazuh during testing. The rest are what FIM is broadly used to catch in a real environment, included here for context rather than claimed as observed.

Tactic	Technique	Notes
Impact	T1565.001 – Stored Data Manipulation	Directly observed: auto-tagged when malware.txt's content was altered
Defense Evasion	T1070.004 – Indicator Removal: File Deletion	Matches the "File deleted" events (rule 553) seen during basic testing
Persistence	T1547.001 – Registry Run Keys / Startup Folder	Matches the monitored Startup folder and registry run-key paths
Defense Evasion	T1112 – Modify Registry	Matches the registry checksum-changed alerts seen throughout testing
Defense Evasion	T1036 – Masquerading	Matches the deliberately suspicious file name (malware-exe) used

6. TROUBLESHOOTING

Picking the right mode

Scheduled, realtime, and who-data modes behave differently, and using the wrong one for a given test wastes time chasing an alert that was never going to include what was being looked for.

Registry noise

Registry monitoring under HKEY_LOCAL_MACHINE generated a steady stream of checksum-changed events, since Windows touches those keys on its own during normal operation. Ignore rules needed tuning so the real test events didn't get lost in that background noise.

Reusing file names across phases

Similar file names (malware, malware-exe) across different test phases made it easy to correlate the wrong dashboard event to the wrong test. Worth being more deliberate about naming next time.

Agent restarts after config changes

Every ossec.conf edit needed a restart of the Wazuh agent service via the Windows Services console before it took effect. Easy to forget, then spend a few minutes wondering why a rule change isn't showing up.

7. TAKEAWAY

Scheduled and realtime FIM alone answers "did something change." Who-data mode plus the MITRE mapping answers who did it, with what process, and how it maps to an actual attack technique. That gap, not just having FIM switched on, is what makes an alert useful in a real investigation instead of just noise on a dashboard.